Erdős Problem 700(iii): Research Map and Open Frontier

Nanocodex Erdős 700 project

3 August 2026

Abstract

This is a status report, not a claimed solution. It records the exact weighted carry reduction for Erdős Problem 700(iii), the strongest retained partial results, the approaches for which we found rigorous counterexamples or quantifier failures, and the remaining same-row synchronization problem. No work in this repository proves the target for any fixed exponent $A > 1$, gives an unbounded improvement over the classical $A = 1$ scale for all composite integers, or constructs a counterexample to the original statement.

1 Target and exact reformulation

For composite $n > 1$, define

$$f(n) = \min_{2 \le k \le \lfloor n/2 \rfloor} \gcd\left(n, \binom{n}{k}\right).$$

Part (iii) asks whether, for every real $A > 0$, there is a constant $C_A > 0$ such that

$$f(n) \le C_A \frac{n}{(\log n)^A} \tag{1}$$

for every composite n. Put

$$D_n(k) = \frac{n}{\gcd(n, \binom{n}{k})}, \qquad D(n) = \max_{2 \le k \le \lfloor n/2 \rfloor} D_n(k).$$

Then (1) is equivalent, after absorbing finitely many small n, to

$$D(n) \ge c_A (\log n)^A \tag{2}$$

for a constant $c_A > 0$ depending only on A.

Write $n = \prod_{p|n} p^{a_p}$ and define the retained depth at a legal row by

$$d_p(k) = \max\left\{a_p - v_p\left(\binom{n}{k}\right), 0\right\}.$$

Primewise valuation gives the exact objective

$$\boxed{\log D_n(k) = \sum_{p|n} d_p(k) \log p.} \tag{3}$$

Kummer identifies the binomial valuation with the number of base-p carries. Thus the open problem is to find one literal legal row k retaining total weighted depth at least

$$A \log\log n - O_A(1). \tag{4}$$

The words “one row” are load-bearing: successes at different primes or scales do not combine unless they occur at the same integer k.

If $q = p^a \parallel n$ and $q \mid k$, scaling gives

$$v_p\left(\binom{n}{k}\right) = v_p\left(\binom{n/q}{k/q}\right), \qquad d_p(k) = \max\left\{a - v_p\left(\binom{n/q}{k/q}\right), 0\right\}. \tag{5}$$

This exact component language underlies all of the retained routes.

2 What has been proved

The following are complete partial results, not substitutes for (1).

1. **Classical and easy regimes.** The greatest-component witness and the classical least-common-multiple estimate recover the $A \le 1$ scale. Prime powers, boundedly many exact prime-power components, a sufficiently large component, and sufficiently large high-height mass are also controlled.

2. **Exact hard-case reduction.** Outside those regimes, the obstruction can be reduced to many comparable, low-height exact prime-power components. Ordering and pigeonholing produce a homogeneous dyadic packet of growing prime bases. The component cap, high-height cutoff, integral-A endpoint, and legal-row endpoint were checked explicitly.

3. **Deep-prefix escape.** For each selected fixed-size subset, explicit Chinese-remainder rows can suppress carries through $\Omega_A(\log n/\log\log n)$ digit levels. Therefore a genuine obstruction must recur in the deep tail; bounded-prefix counterexamples cannot settle the problem.

4. **Exact effective conductor.** For

$$\mathcal{A} = \{0 \le z < p^D : z \le T, v_p\binom{T}{z} < s\}, \qquad \Delta = p^D - T,$$

   the least residue conductor retained by the campaign is

$$E = \begin{cases} 0, & \Delta = 1, \\ \min\bigl(D, \lceil \log_p \Delta \rceil + s - 1\bigr), & \Delta > 1. \end{cases} \tag{6}$$

   This replaces an abstract tail period by an exact top-gap statistic.

5. **Limits on structured blockers.** Exact power-word anchors in a short prime packet have pairwise-coprime exponents, yielding a sub-packet upper bound on their number. A separate cyclotomic argument excludes one sparse four-digit endpoint architecture for fixed $A < 2$. Neither result classifies all hostile digit words.

6. **Conditional transfer.** Any construction of one legal row satisfying (4) proves Part (iii). The often-used upper-half first-layer condition is a stronger sufficient condition, not an equivalent reformulation.

3 Attempt map and no-retry lessons

Route family	Disposition and retained lesson
Fixed multiplier menus and naive witness iteration	Explicit counterexamples show that no fixed finite menu, copied component row, or local bounded-loss peeling rule preserves enough depth uniformly.
Independent carry probabilities and abstract set cover	Marginal success at separate primes does not produce one common row. Any averaging proof needs arithmetic correlation in the actual signed phases.
Pair and packet alignment	Several universal pair-selection and upper-half full-layer statements are false. An instance-adaptive multiplier or an inverse theorem for hostile words remains plausible.
Carmichael, lcm, determinant, and resultant bridges	Useful exact identities survive, but a fixed cubic Carmichael bridge is false and bounded-degree algebra loses the growing digit tail.
Digit boxes and affine rigidity	Exact escape bounds and sharp examples were proved. Total box entropy or active-digit count alone cannot force a positive fraction of useful multipliers.
Large sieve, characters, and moment methods	Taking absolute values destroys the phase needed for simultaneous moving-base control. Fixed moments stop at the singleton/$A = 1$ scale.
Counterexample construction	We found counterexamples to methods, not to Erdős 700(iii). No infinite family with a uniform all-row bound contradicting (2) is known here.

These routes should not be restarted unchanged. A new version must state the additional arithmetic input that defeats the recorded counterexample or repairs the failed quantifier.

4 The surviving bottleneck

The most direct target is a weighted-depth preservation theorem: move inside an explicit legal progression so as to gain a new prime-power layer while losing only a controlled fraction of previously retained weight. A finite iteration would need to reach (4).

For $1 < A \leq 2$, the residual packet is essentially squarefree, so extra prime-power height gives no slack. The central pair problem is therefore:

> Given the actual common cofactor and a sufficiently large packet of primes, find distinct p, q and one legal multiplier satisfying both complete base-p and base-q Lucas digit inequalities.

A plausible route is an inverse theorem: long affine avoidance of a Lucas digit down-set forces the upper word into a sparse or near-power class; a moving-base separation theorem would then show that only a negligible fraction of packet primes can be hostile for the same ambient integer. Alternatively, an all-row contradiction could average the exact quantities in (3), but it must retain their signed cross-base phase instead of multiplying marginal probabilities.

5 What would count as progress

The repository's acceptance gate for Part (iii) is intentionally concrete. Any one of the following would materially move the frontier:

1. a uniform lower bound $D(n) \geq L(n) \log n$ with $L(n) \to \infty$;
2. a proof of (2) for $A = 2$;
3. the full theorem for every fixed $A > 0$;
4. an infinite counterfamily negating (2) for one fixed $A > 0$;

5. a carry-preserving composition or moving-base inverse theorem strong enough to imply (4).

6 Evidence and source trail

The detailed theorem-by-theorem ledger, including the named no-retry constraints and the input required to revive each route, is `map/part-3.md`. The compact specialist handoff is `map/part-3-bottleneck.md`. Persisted-run and session coverage audits are linked from the repository research map.

The primary historical source is P. Erdős and G. Szekeres, “Some number theoretic problems on binomial coefficients,” *Australian Mathematical Society Gazette* 5 (1978), 97–99, archival PDF.